Year 9 Science Test 1 Core paper

Biology (40%): variation, adaptation, natural selection, cells · Chemistry (20%): states of matter, atoms · Physics (40%)

DRAFT Not ready to print for students: Part 3: Physics (16 marks) has not been added yet.

Name	Class	Date	Score / 40

Instructions · Time: 40 minutes · Total: 40 marks

- You have 40 minutes. There are three parts: Biology (16 marks), Chemistry (8 marks) and Physics (16 marks). Answer all the questions.
- Spend about one minute on each mark: 16 minutes on Biology, 8 minutes on Chemistry and 16 minutes on Physics.
- The last question in each part is harder (Stretch and Challenge). Answer the other questions in that part first.
- Write your answers in the spaces on this paper. The marks for each part of a question are shown in brackets, for example [2].

Key words 关键词

abiotic factor	非生物因素	**biotic factor**	生物因素	**continuous variation**	连续变异
genetic information	遗传信息	**nucleus**	细胞核	**fertilisation**	受精
zygote	受精卵	**nature reserve**	自然保护区	**breeding programme**	繁育计划
gene bank	基因库	**natural selection**	自然选择	**gene**	基因
generation	世代	**camouflage**	伪装	**cell wall**	细胞壁
chloroplast	叶绿体	**vacuole**	液泡	**magnification**	放大倍数
eyepiece lens	目镜	**objective lens**	物镜	**particle**	粒子
solid	固体	**liquid**	液体	**gas**	气体
proton	质子	**neutron**	中子	**electron**	电子
atomic number	原子序数	**mass number**	质量数	**electron configuration**	电子排布
periodic table	元素周期表	**group**	族		

Command words 指令词: Calculate 计算 · Complete 完成（填写） · Draw 画出 · Explain 解释 · Predict 预测 · Tick 打勾 · Use 使用（运用） · Write 写出

Part 1: Biology

16 marks · about 16 minutes

Questions 1 to 5 are Core. Question 6 is Stretch and Challenge.

1 Environmental variation

(3 marks)

(a) Put one tick in each row to show whether the factor is abiotic (non-living) or biotic (living). **[2]**

Factor	Abiotic	Biotic
temperature		
predation (being hunted)		
amount of light		
infectious disease		

(b) Which one of these shows continuous variation? Tick one box. **[1]**

- ☐ blood group
- ☐ height
- ☐ tongue rolling (can or cannot)
- ☐ natural eye colour

2 Inherited variation

(2 marks)

(a) Complete the sentences using words from the box. One word is not needed. **[2]**

nucleus	**fertilisation**	**environment**

Genetic information is stored in the _____ of a cell.

A sperm cell and an egg cell join together at _____ to make a zygote.

3 Adaptation and extinction

(2 marks)

(a) Draw one line from each way of protecting a species to how it works. **[2]**

nature reserve	endangered animals are bred in captivity to increase their numbers
breeding programme	seeds, gametes or cells are stored at a low temperature
gene bank	an area where the habitat and the species living in it are protected

4 Natural selection

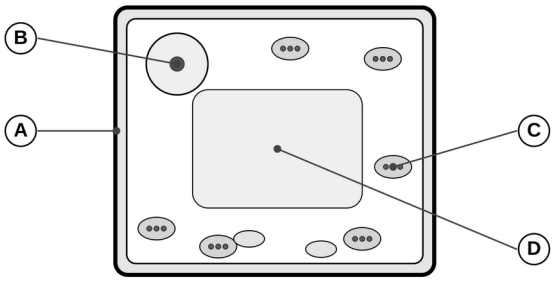
(2 marks)

(a) In the 1800s, black peppered moths became more common near Manchester. The sentences below explain why, but they are in the wrong order. Write 1, 2, 3 and 4 in the boxes to put them in the correct order. **[2]**

Order	Sentence
	Birds spotted and ate more pale moths than black moths.
	A few moths had genes that made them black.
	Each generation had more black moths than the one before.
	Soot from factories made the tree trunks black.

5 Cells and the microscope (3 marks)

(a) The diagram shows a plant cell. Write the name of each labelled part. **[2]**



A plant cell seen through a microscope (not to scale)

A _____

B _____

C _____

D _____

(b) A microscope has a x10 eyepiece lens and a x4 objective lens. Calculate the total magnification. **[1]**

Total magnification = x _____

6 Peppered moths today STRETCH AND CHALLENGE (4 marks)

(a) By 1895 most peppered moths near Manchester were black, because soot from factories had made the tree trunks black. Since the 1960s the air has been much cleaner. There is very little soot, and the tree trunks are pale again. **[4]**

(i) Predict what has happened to the numbers of black moths and pale moths near Manchester since the 1960s.

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(ii) Explain your prediction. Use the idea of natural selection.

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Part 2: Chemistry

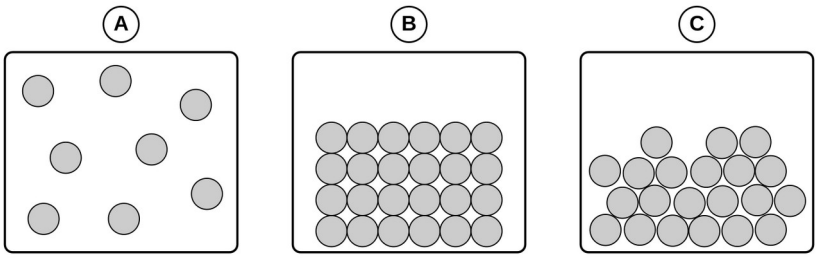
8 marks · about 8 minutes

Questions 7 and 8 are Core. Question 9 is Stretch and Challenge.

7 States of matter

(2 marks)

(a) The diagrams show the particles in a solid, a liquid and a gas. Write solid, liquid or gas for each diagram. **[2]**



A _____

B _____

C _____

8 Atoms and the periodic table

(4 marks)

(a) A sodium atom has atomic number 11 and mass number 23. **[2]**

(i) How many protons does the atom have?

protons = _____

(ii) How many neutrons does the atom have?

neutrons = _____

(b) A sodium atom has 11 electrons. Complete its electron configuration. **[1]**

2, 8, _____

(c) Use your answer to part (b). Which group of the periodic table is sodium in? Tick one box. **[1]**

- ☐ group 1
- ☐ group 2
- ☐ group 7
- ☐ group 0

9 Elements in the same group STRETCH AND CHALLENGE

(2 marks)

(a) Fluorine (atomic number 9) and chlorine (atomic number 17) are both in group 7 of the periodic table. **[2]**

(i) Write the electron configuration of each atom.

fluorine _____ chlorine _____

(ii) Explain why fluorine and chlorine react in similar ways.

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Part 3: Physics

16 marks · about 16 minutes

Questions from 10 onwards. The last question will be Stretch and Challenge.

Physics questions to be added: 16 marks, about 16 minutes. Planned as 12 marks of Core questions and one 4-mark Stretch and Challenge question, matching Part 1. The topics taught in physics are needed before these can be written.

END OF TEST